

Comparative Performance Analysis of MILP Solvers for Cryptanalysis

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Abstract. This paper provides a performance comparison of five MILP solvers applied to related-key differential cryptanalysis of ITUbee [7]. We evaluate three open-source solvers (GLPK, HiGHS, SCIP) and two commercial solvers (Gurobi, CPLEX) using MILP models for 8, 10, and 12-round attacks. As rounds increase, the number of equations and constraints grows exponentially. Experiments used an 11th Gen Intel Core i7-1165G7 processor with 32 GB of RAM. Commercial solvers (Gurobi and CPLEX) perform better than open-source options, achieving up to 94× speedup compared to GLPK for the 12-round model. This work provides guidance for choosing a solver for MILP-based cryptanalysis.

Keywords: Cryptanalysis · MILP · Differential Cryptanalysis · Block Ciphers

1 Introduction

MILP has become a standard tool for automated cryptanalysis of block ciphers, especially when searching for optimal differential and related-key characteristics. In this setting, the choice of MILP solver often determines whether an analysis is practical or infeasible, since solver runtime can vary by orders of magnitude on the same model.

This paper focuses on the performance comparison of five MILP solvers on related-key differential models for the lightweight cipher ITUbee. We benchmark three open-source solvers (GLPK, HiGHS, SCIP) and two commercial solvers (Gurobi, CPLEX) on 8-, 10-, and 12-round instances. The comparison highlights how algorithmic design, preprocessing, and search heuristics affect solver efficiency on cryptanalytic MILP workloads.

ITUbee is a software-oriented lightweight Feistel cipher operating on 80-bit blocks and an 80-bit key [7]. Its round structure includes key whitening, substitution, and linear diffusion layers, making it a suitable target for MILP-based related-key differential analysis. This paper uses ITUbee as a representative case study for solver benchmarking rather than as a proposal for a new cipher design.

The main contributions of this work are:

- a direct empirical comparison of GLPK, HiGHS, SCIP, Gurobi, and CPLEX on a series of ITUbee-related MILP models;

- an analysis of how solver performance scales as the model difficulty increases from 8 to 12 rounds;
- practical recommendations for choosing a solver for MILP-based related-key differential cryptanalysis.

The results are intended to guide cryptanalysts who need to select the most effective solver for MILP-based analysis, with a clear emphasis on performance rather than cryptographic design.

Organization of the paper. Section 2 introduces MILP-based cryptanalysis and related-key differential attacks. Section 3 describes the ITUbee cipher and the MILP model. Section 4 presents the benchmark results and analysis. Section 5 concludes with practical solver recommendations.

2 Our MILP Model

In this study, we adopt the related-key differential cryptanalysis model introduced in [6]. To facilitate a comprehensive comparison, 8-, 10-, and 12-round instances of the model are employed to evaluate the performance of various solvers. For the sake of completeness, a brief overview of related-key differential cryptanalysis and the ITUbee cipher is also provided; for a more detailed exposition, the reader is referred to [6].

2.1 ITUbee

ITUbee [7] is a lightweight software-oriented block cipher specifically designed for resource-constrained devices which include a microcontroller and have a limited battery power such as sensor nodes in wireless sensor networks.

The cipher has Feistel structure which consists of 20 rounds. It has 80 bits block size and 80 bits key size. The round structure of the cipher is given in Figure 1. Cipher also has a key whitening layer in the beginning and end of the 20 rounds. The encryption process of the cipher is given in Algorithm 1.

L Function The L function which has branch number 4 is computed as,
 $L(a \parallel b \parallel c \parallel d \parallel e) = (e \oplus a \oplus b) \parallel (a \oplus b \oplus c) \parallel (b \oplus c \oplus d) \parallel (c \oplus d \oplus e) \parallel (d \oplus e \oplus a)$

F Function The F function is computed as,

$$F(X) = S(L(S(X)))$$

S is the S-box used in AES [10]. S is the only non-linear structure in the ITUbee algorithm. 40 bit input of S is splitted into 8 bits then enters the s-box.

$$S(a \parallel b \parallel c \parallel d \parallel e) = s[a] \parallel s[b] \parallel s[c] \parallel s[d] \parallel s[e]$$

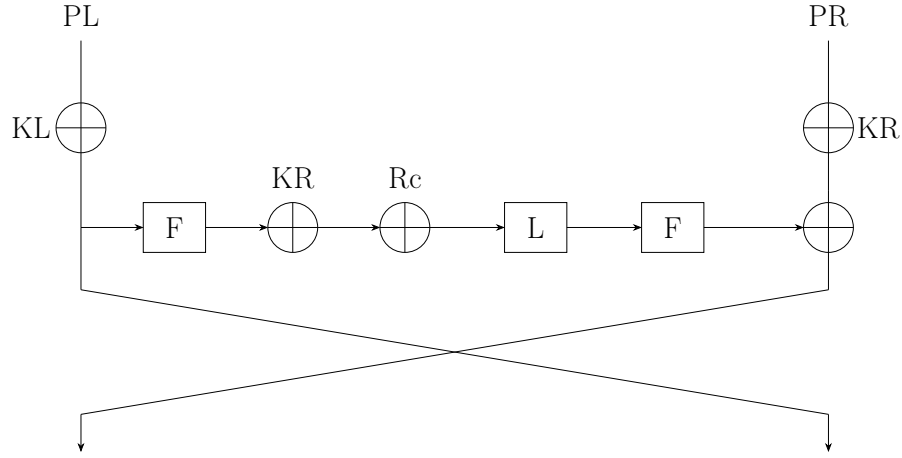


Fig. 1. ITUbee round structure

Algorithm 1 Pseudo code for ITUbee encryption

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1:  $X_1 \leftarrow P_L \oplus K_L$ 
2:  $X_0 \leftarrow P_R \oplus K_R$ 
3: for  $i = 1$  to 20 do
4:   if  $i \in \{1, 3, \dots, 19\}$  then
5:      $RK \leftarrow K_R$ 
6:   else
7:      $RK \leftarrow K_L$ 
8:   end if
9:    $X_{i+1} \leftarrow X_{i-1} \oplus F(L(RK \oplus RC_i \oplus F(X_i)))$ 
10:  Round constant is added to the rightmost 16 bits
11: end for
12:  $C_L \leftarrow X_{20} \oplus K_R$ 
13:  $C_R \leftarrow X_{21} \oplus K_L$ 

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Key Schedule ITUbee has no key schedule. The algorithm splits the 80 bit key into two 40 bit parts K_L and K_R and uses the right part of the key in odd rounds and the left part of the keys in even rounds.

2.2 Related-key Differential Cryptanalysis

For the related-key model, we must explicitly represent the key differences in addition to the data differences. To do this, we introduce binary variables for key activity and add constraints that force at least one key byte to be active when a related-key difference is present. Without these constraints, the model could avoid key activity and would effectively reduce to the ordinary differential model.

This related-key model also contains more intermediate variables than the differential and linear models, because we must represent the extra variables that

appear after the key XOR operation in the middle of the round. The modeling of XOR, linear, and key-dependent components follows the MILP approach of Mouha and Preneel [8]. A sketch of the one-round related-key MILP model is shown in Figure 2.

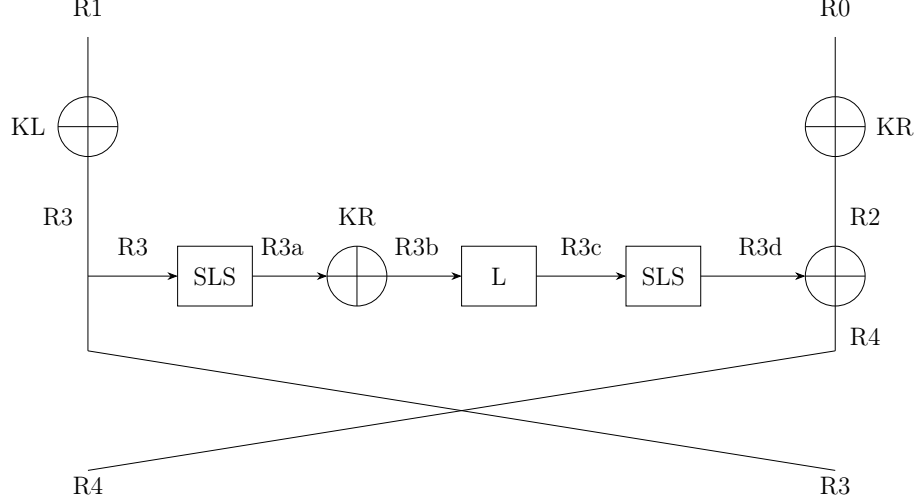


Fig. 2. ITUbee MILP Related-Key Differential Cryptanalysis Sketch

We found that 8 round of the algorithm has minimum 16 active s-box. ITUbee algorithm uses AES's [2] s-box. AES s-box have best input-output difference 2^{-6} . Therefore best probability for differential characteristic will be $(2^{-6})^{16} = 2^{-96}$ which is not usable for related key differential attack. Authors of the ITUbee algorithm claims in [7] that algorithm is secure against related key differential cryptanalysis after 10 round. In our model we found that 8 round is enough for security against related key differential cryptanalysis. Best 8 round characteristic found with the model can be seen in Table 1

3 Solver Specifications

This section provides detailed descriptions of the five MILP solvers evaluated in our comparative study.

3.1 GLPK (GNU Linear Programming Kit)

GLPK is a free open-source linear programming solver developed as part of the GNU project [9]. Version 4.65 was used in our experiments. GLPK uses the revised simplex method for linear programming and branch-and-cut algorithms for

integer constraints. While GLPK provides reliable and deterministic solutions, it does not include the same heuristic or parallel features as commercial solvers, so it is slower for larger MILP instances. GLPK is widely used in academic research because it is accessible and performs well on small to medium-sized problems.

3.2 HiGHS (High-performance Interior-point Solver)

HiGHS (version 0.9) is an open-source linear and mixed-integer programming solver developed by the University of Edinburgh [3]. It uses simplex, interior-point, and active-set methods for LP problems, together with parallel branch-and-bound techniques for MILP. HiGHS is designed to be efficient and supports parallel execution. In practice, HiGHS is often faster than GLPK on many problem instances, especially when using multiple cores.

3.3 SCIP (Solving Constraint Integer Programs)

SCIP (version 7.0) is an open-source constraint integer programming framework developed by the Zuse Institute Berlin [1]. Beyond basic MILP, SCIP supports mixed-integer nonlinear programming and quadratically constrained problems. SCIP uses cutting planes, primal heuristics, and symmetry breaking. Among open-source solvers, SCIP often performs well and can be a strong option for researchers working with harder problem classes.

3.4 Gurobi Optimizer

Gurobi (version 10.0) is a commercial MILP solver developed by Gurobi Optimization [4]. It uses barrier methods (interior-point), parallel branch-and-cut, and presolve techniques. Gurobi also includes primal and dual heuristics and can distribute work across multiple processor cores. Its tuning and parameter options allow control over the solving process. It is widely used in industry and is a common benchmark for commercial solvers.

3.5 CPLEX (IBM CPLEX Optimization Studio)

CPLEX (version 12.10) is IBM's commercial optimization solver with decades of development [5]. It uses primal and dual simplex, barrier methods with crossover, and branch-and-cut methods. CPLEX supports parallel processing and includes preprocessing and probing techniques. It is widely used in enterprise optimization and is one of the main commercial options for MILP solving.

3.6 Experimental Configuration

All solvers were executed with default parameters to ensure fair and reproducible comparison. This approach reflects typical usage patterns where practitioners rely on default settings without extensive parameter tuning. Each solver was

allowed unlimited computation time (no time limits were imposed) to find optimal solutions. Experiments were conducted on an 11th Gen Intel Core i7-1165G7 processor operating at 2.80 GHz with 32 GB of system RAM, running Ubuntu Linux. This ensures all solvers had equal access to computational resources, with no artificial constraints or advantages.

4 Results and Analysis

Comprehensive benchmarking was conducted on MILP models for 8, 10, and 12-round related-key differential attacks on ITUbee. The results demonstrate clear performance distinctions between solver categories, with significant implications for practical related-key differential cryptanalysis.

4.1 8-Round Results

Our 8-round model consists of 687 variables and 2147 constraints. Our results show that GLPK, the baseline open-source solver, required 257 seconds to find a solution. HiGHS improved on this with a time of 231 seconds, while SCIP further reduced the time to 86 seconds. The commercial solvers Gurobi and CPLEX achieved faster times of 37 and 16 seconds, respectively, representing speedups of $6.9\times$ and $16.0\times$ compared to GLPK (see Table 2).

Table 2. Solution times for the 8-round model

Solver	Time (s)	Speedup
GLPK	257	$1.0\times$
HiGHS	231	$1.1\times$
SCIP	86	$2.9\times$
Gurobi	37	$6.9\times$
CPLEX	16	$16.0\times$

4.2 10-Round Results

The 10-round model consists of 945 variables and 3182 constraints. In this harder instance, GLPK required 1610 seconds to solve the problem. HiGHS improved substantially, finishing in 321 seconds, and SCIP further reduced the time to 202 seconds. The commercial solvers again delivered the best performance, with Gurobi solving the model in 34 seconds and CPLEX in 40 seconds, representing speedups of $47.4\times$ and $40.3\times$ versus GLPK, respectively (see Table 3).

Table 3. Solution times for the 10-round model

Solver	Time (s)	Speedup
GLPK	1610	1.0×
HiGHS	321	5.0×
SCIP	202	7.9×
Gurobi	34	47.4×
CPLEX	40	40.3×

4.3 12-Round Results

The 12-round model is the most challenging benchmark, with 1203 variables and 4217 constraints. GLPK required 4543 seconds to solve this instance, while HiGHS completed in 1780 seconds. SCIP again offered a strong open-source alternative at 376 seconds. Gurobi and CPLEX achieved the fastest times of 48 and 53 seconds, corresponding to speedups of 94.6× and 85.7× relative to GLPK (Table 4).

Table 4. Solution times for the 12-round model

Solver	Time (s)	Speedup
GLPK	4543	1.0×
HiGHS	1780	2.5×
SCIP	376	12.0×
Gurobi	48	94.6×
CPLEX	53	85.7×

4.4 Discussion

The experimental results reveal several important trends:

1. **Commercial vs. Open-Source Gap:** Commercial solvers (Gurobi and CPLEX) consistently outperform open-source alternatives, with speedups increasing dramatically for larger problem sizes. For the 12-round model, Gurobi achieves a 94.6× speedup over GLPK.
2. **Problem Scaling:** As the problem complexity increases (8 to 12 rounds), the performance gap between solvers widens. This suggests that the commercial solvers use methods that are more effective on harder instances.
3. **Open-Source Alternatives:** Among open-source solvers, SCIP consistently outperforms both GLPK and HiGHS, achieving 2.9× speedup on 8-rounds and 12× speedup on 12-rounds compared to GLPK.
4. **Commercial Solver Parity:** Gurobi and CPLEX show comparable performance, with Gurobi slightly faster on 10 and 12-round models, while CPLEX excels on the 8-round problem.

The timing analysis demonstrates that solver choice is critical for practical cryptanalysis. For resource-constrained deployments, SCIP provides a reasonable alternative to commercial solvers. However, for optimal performance, commercial solutions are strongly recommended.

5 Conclusion

This paper provides a comparison of five MILP solvers in the context of cryptanalytic applications, specifically related-key differential attacks on the ITUbee lightweight block cipher. Our findings show that solver selection has a large effect on the feasibility of MILP-based related-key differential cryptanalysis.

Key conclusions include:

1. Commercial MILP solvers (Gurobi and CPLEX) are more efficient than open-source alternatives and can be preferred for larger, time-sensitive analyses.
2. For organizations without access to commercial solvers, SCIP is a good open-source option that performs well on moderate problem sizes.
3. The performance advantage of commercial solvers grows with problem complexity, suggesting that their methods are more effective on harder instances.
4. Cryptanalysts should evaluate solver options based on their constraints (computational budget, licensing availability, problem size) when conducting related-key differential cryptanalysis.

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Stage	Binary Index	Characteristic
Key	KL	{0: 0.0, 1: 1.0, 2: 0.0, 3: 1.0, 4: 0.0}
	KR	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
1. round input	R1	{0: 0.0, 1: 1.0, 2: 0.0, 3: 1.0, 4: 0.0}
	R0	{0: 1.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 1.0}
1. round middle values	R2	{0: 1.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 1.0}
	R3-3a-3b-3c-3d	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
2. round input	R4	{0: 1.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 1.0}
	R3	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
2. round middle values	R4a	{0: 0.0, 1: 1.0, 2: 0.0, 3: 1.0, 4: 0.0}
	R4b-4c-4d	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
3. round input	R5	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
	R4	{0: 1.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 1.0}
3. round middle values	R5a-5b-5c-5d	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
4. round input	R6	{0: 1.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 1.0}
	R5	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
4. round middle values	R6a	{0: 0.0, 1: 1.0, 2: 0.0, 3: 1.0, 4: 0.0}
	R6b-6c-6d	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
5. round input	R7	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
	R6	{0: 1.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 1.0}
5. round middle values	R7a	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
	R7b-7c-7d	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
6. round input	R8	{0: 1.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 1.0}
	R7	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
6. round middle values	R8a	{0: 0.0, 1: 1.0, 2: 0.0, 3: 1.0, 4: 0.0}
	R8b-8c-8d	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
7. round input	R9	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
	R8	{0: 1.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 1.0}
7. round middle values	R9a-9b-9c-9d	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
8. round input	R10	{0: 1.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 1.0}
	R9	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
8. round middle values	R10a	{0: 0.0, 1: 1.0, 2: 0.0, 3: 1.0, 4: 0.0}
	R10b-10c-10d	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
9. round input	R11	{0: 0.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 0.0}
	R10	{0: 1.0, 1: 0.0, 2: 0.0, 3: 0.0, 4: 1.0}

Table 1. Best 8 round related-key differential characteristic of ITUbee algorithm